Final design

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WTE plant size(KW) =25000, Wind Turbines =2.924439e+03, Solar Power(KW) =4.447824e+04, Battery units =3.137920e+04

Total Wind energy(KWh)=

4.4827e+06

Total Solar energy(KWh)=

7.3256e+07

Total DG Energy generation(KWh)=

0

Total WTE Energy generation(KWh)=

6.3714e+07

Wasted Energy(KWh)=

3.1552e+07

Total Load demand(KWh)=

1.0715e+08

LCOE_ =

0.2209

REF =

100

WT_fraction =

5.7664

PV_fraction =

94.2336

Co2_Emission =

5.7343e+10

TNPC =

3.8437e+08

****

Annualized cost of components

****

Anual_cost =

2.3667e+07

Wind_cost =

1.5631e+07

WTE_cost =

1.8832e+08

Solar_cost =

1.0284e+08

DG_cost =

0

Battery_cost =

7.7539e+07

Inverter_Tcost =

4.1373e+04

Bin =

5.3147e+06

Bout =

4.8315e+06